# 1.4 Matrix Group

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Matrix Group

Notation : In $\mathbb{C}^n$, $\overline{e_i} = (0, \cdots, i^{\text{th}}, 0 \cdots)$ is a vector space basis element.

* An endomorphism $\alpha$ of $\mathbb{C}^n$ is defined as

$$\alpha\overline{e_i} = \sum_{j=1}^{n} a_{ji}\overline{e_j}$$

where $\alpha = (a_{ij}) \in M_n(\mathbb{C})$.

(We can think of $\alpha\overline{e_i}$ as the $i^{\text{th}}$ column vector.)

We use $\alpha$ for a matrix as well as for the endomorphism.

We define multiplication of two endomorphisms $\alpha, \beta$ with corr. matrices $(a_{ij}), (b_{ij})$ and construct $(c_{ij})$ where

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

**Lemma 0.1.** $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ as vector spaces.

*Proof.* Construct the function

$$\tilde{f} : M_n(\mathbb{C}) \to \mathbb{C}^{n^2}$$
$$(a_{ij}) \mapsto (b_{ij})$$

where $b_{i+(j-1)n} = a_{ij}$.

Observe as $j$ goes from 1 to $n$, $(j-1)n$ goes from 0 to $(n-1)n$ and as $i$ goes from 1 to $n$ as well, we get $i + (j-1)n$ go from 1 to $n^2$.

Observe $\tilde{f}$ is surjective by construction and if $a_{ij} \neq a_{i'j'}$ then $b_{i+(j-1)n} \neq b_{i'+(j'-1)n}$ hence $\tilde{f}$ is bijective.

Observe $\tilde{f}$ is linear hence $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$. $\square$

*Remark* 0.1. Observe we can equip $M_n(\mathbb{C})$ with a topology by ensuring the given map above to be a homeomorphism. By constructing

$$\tau_{M_n(\mathbb{C})} = \{f^{-1}(U) \mid U \in \tau_{\mathbb{C}^{n^2}}\}$$

**Lemma 0.2.** Let $T$ be any topological space,

$$\phi : T \to M_n(\mathbb{C})$$

$\phi$ is continuous if and only if $\pi_{ij} \circ \phi$ is continuous.

*Proof.* Only if condition is trivial.

If condition:

$$\pi_{ij} \circ \phi : T \to \mathbb{C} \text{ is continuous.}$$

Now $\prod_{1 \le i,j \le n} \pi_{ij} \circ \phi : T \to \mathbb{C}^{n^2}$ is continuous by product topology (or box topology as $\mathbb{C}^{n^2}$ is finite Cartesian product). □

$$\text{as } \pi_{ij}^{-1}(U) \text{ for } U \text{ open in } \mathbb{C} \text{ forms a subbasis for product topology on } \mathbb{C}^{n^2}.$$

□

**Corollary 0.1.** Product of two matrices is continuous

*Proof.*

$$\text{Pf: } M_n(\mathbb{C}) \times M_n(\mathbb{C}) \xrightarrow{q} M_n(\mathbb{C})$$

$$f \times f$$
$$\mathbb{C}^{n^2} \times \mathbb{C}^{n^2} \quad \xrightarrow{p} \quad \mathbb{C}^{n^2}$$

where $q : M_n(\mathbb{C}) \times M_n(\mathbb{C}) \to M_n(\mathbb{C})$

$$(\sigma, \tau) \mapsto \sigma\tau$$

is matrix multiplication. And $p$ is a polynomial to $p((b_{ij}), (b'_{ij})) = (c_{ij})$ where

$$c_{ij} = \sum_{k=1}^{n} b_{ik} b'_{kj}.$$

Hence $p$ is continuous. Therefore for $q = p \circ (f \times f)$ is continuous. For any open set in $M_n(\mathbb{C})$ $U$, we express $U = f^{-1}(O)$ for some open set $O$ in □

$$\mathbb{C}^{n^2}.$$

Hence $q^{-1}(U) = q^{-1}(f^{-1}(O)) = (f \circ q)^{-1}(O)$, which is open as $f \circ q$ is continuous.

□

Notation: ${}^tA$ is transpose of a matrix and $\overline{A}$ is complex conjugate of a matrix.

**Lemma 0.3.** Transpose and complex conjugate are homeomorphism.

*Proof.* If $h : M_n(\mathbb{C}) \to M_n(\mathbb{C})$,

$$A \xrightarrow{h} {}^tA$$

we can construct $r : \mathbb{C}^{n^2} \to \mathbb{C}^{n^2}$ which is a rearrangement of indices such that

$$b_{i+j-n} = b_{j+i-c-1-n} \quad \text{which is a homeomorphism.}$$

Hence, $h = f^{-1}rf$ is a homeomorphism.

If $k : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ is homeomorphic

$$A \xrightarrow{k} \overline{A}$$

as $k_{ij}$ is homeomorphic for $1 \le i, j \le n$.

Property of matrices

$$(\alpha\beta)^t = \beta^t\alpha^t$$
$$\frac{d(\alpha\beta)}{dt} = \dot{\alpha}\beta + \alpha\dot{\beta}$$

* A regular/non singular $n \times n$ matrix, if it has an inverse i.e. if there exists a matrix $\sigma^{-1}$ such that $\sigma\sigma^{-1} = \sigma^{-1}\sigma = E$, where $E$ is the unit matrix of degree $n$.

A necessary and sufficient condition for $\sigma$ to be regular is $\det \sigma \neq 0$.

*Remark* 0.2. If $\sigma$ is a regular matrix

(i) iff $\sigma^{-1}$ its reciprocal endomorphism exists

(ii) then ${}^t(\sigma^{-1}) = ({}^t\sigma)^{-1}$, $(\bar{\sigma})^{-1} = (\bar{\sigma^{-1}})$

*Remark* 0.3. If $\sigma, \tau$ are regular matrices then $\sigma\tau$ are regular matrices.

$*$ $\det : M_n(\mathbb{C}) \to \mathbb{C}$ can be expressed as

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\det} & \mathbb{C} \\ \downarrow f & & \\ \mathbb{C}^{n^2} & \xrightarrow{p} & \mathbb{C} \end{array}$$

$p$ is some polynomial function of $n^2$ variables, which is continuous. We know $f$ is continuous. Hence, det map is continuous.

*Remark* 0.4. $GL_n(\mathbb{C}) = \{\sigma \in M_n(\mathbb{C}) \mid \det \sigma \neq 0\}$

$= M_n(\mathbb{C}) \setminus \det^{-1}\{0\}$

is a group and an open subset as det is a continuous map.

(ii) For $\sigma = (a_{ij}) \in GL_n(\mathbb{C})$ the coefficient $b_{ij}$ of $\sigma^{-1}$ are given by $b_{ij} = \frac{1}{\det \sigma} A_{ij}$ where $A_{ij}$ is a polynomial of $\sigma$. Hence mapping $\sigma \mapsto \sigma^{-1}$ is continuous on $GL(n, \mathbb{C})$. Moreover, it's a homeomorphism.

*Remark* 0.5. If $\sigma \in GL_n(\mathbb{C})$, we define $\sigma^* = ({}^t\sigma)^{-1}$. Then we have $(\sigma\tau)^* = \sigma^*\tau^*$. Hence $\sigma \mapsto \sigma^*$ is a homeomorphism, and an automorphism of order 2 of $GL_n(\mathbb{C})$.

**Definition 0.1.** $\mathrm{O}_n(\mathbb{C}) = \left\{\sigma \in \mathrm{GL}_n(\mathbb{C}) \mid \sigma^* = \sigma^{-1}\right\}$

is set of orthogonal matrices. If $\sigma = \sigma^*$ then $\sigma$ is complex orthogonal and if $\sigma = \sigma^* = \overline{\sigma}$ then real orthogonal.

$\mathrm{U}_n(\mathbb{C}) = \left\{\sigma \in \mathrm{GL}_n(\mathbb{C}) \mid \sigma^* = \overline{\sigma}^{-1}\right\}$

is set of all unitary matrices.

Note $\mathrm{U}_n(\mathbb{C})$ or $\mathrm{O}_n(\mathbb{C})$ are non empty as $\mathbb{E} \in \mathrm{U}_n(\mathbb{C})$ and $\mathbb{E} \in \mathrm{O}_n(\mathbb{C})$.

Notation: $\mathrm{U}_n(\mathbb{C}) = (\mathrm{UC}_n)$ and $\mathrm{O}_n(\mathbb{R}) = \mathrm{OC}_n$.

Don't be confused!

**Lemma 0.4.** If $f, g$ are continuous mapping from topological space $X$ to Hausdorff space $Y$ then

$$\{x \in X \mid f(x) = g(x)\}$$

is closed.

*Proof.* Consider $\{x \mid f(x) \neq g(x)\}$. Now

$$x \in \{x \in X \mid f(x) \neq g(x)\}$$

implies $f(x) \neq g(x)$ in $Y$. By Hausdorff property there exists open set $U_1$ and $U_2$ such that $f(x) \in U_1$ and $g(x) \in U_2$ and $U_1 \cap U_2 = \emptyset$. Now $f^{-1}(U_1)$ and $g^{-1}(U_2)$ are open. Let $V = f^{-1}(U_1) \cap g^{-1}(U_2)$ be a neighborhood of $x$. For any $y \in V$ $f(y) \neq g(y)$ otherwise $U_1 \cap U_2 \neq \emptyset$. Hence $x \in V \subseteq \{x \mid f(x) \neq g(x)\}$ is open. Hence $\{x \mid f(x) = g(x)\}$ is closed. □

**Proposition 0.1.** $O_n(\mathbb{C})$, $U(n)$, $O(n)$ are closed subgroups of $GL_n(\mathbb{C})$.

*Proof.* Observe mapping $\sigma \to \sigma^*$, $\sigma \to \overline{\sigma}$ and identity mapping are continuous. By the above lemma, as $\mathbb{R}$ is $T_2$ space. The sets ${}_{n \leq \sigma | \overline{\sigma} = \sigma^*}$ and ${}_{n \leq \sigma | \sigma = \sigma^*}$ are closed sets. Observe $GL_n(\mathbb{R}) = \{\sigma | \sigma = \overline{\sigma}\}$ is also closed. Hence $U(n)$ and

$$O(n) = O_n(\mathbb{C}) \cap GL_n(\mathbb{R})$$

is closed. As all the given sets are non-empty by subgroup test, we can show these are subgroups. □

$$* \quad SL_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \det \sigma = 1\}$$

is special linear group. As $\epsilon \in SL_n(\mathbb{C})$, and by subgroup tests we can show $SL_n(\mathbb{C})$ is a subgroup of $GL_n(\mathbb{C})$. Moreover, $SL_n(\mathbb{C}) = \det^{-1}\{1\}$ implies $SL_n(\mathbb{C})$ is a closed subgroup.

*Remark* 0.6.

$$SL(n) = GL_n(\mathbb{C}\mathbb{I}\mathbb{R}) \cap SL_n(\mathbb{C})$$

$$SU(n) = SL_n(\mathbb{C}) \cap U(n)$$

$$SO(n) = SL_n(\mathbb{C}) \cap O(n)$$

are closed subgroups of $GL_n(\mathbb{C})$.

*Theorem* 0.1. $U(n)$, $O(n)$, $SU(n)$ and $SO(n)$ are compact.

*Proof.* Observe $O(n)$, $SU(n)$ and $SO(n)$ are closed subgroups of $U(n)$. It is suffice to show $U(n)$ is compact.

Goal: $U(n)$ is homeomorphic to a closed and bounded set of $\mathbb{C}^{n^2}$.

Observe $GL_n(\mathbb{C})$ is a open subgroup of $M_n(\mathbb{C})$ hence it is also a closed subgroup according to a previous lemma. Observe $U(n)$ is closed in $GL_n(\mathbb{C})$ and $GL_n(\mathbb{C})$ is closed in $M_n(\mathbb{C})$ implies $U(n)$ is closed in $GL_n(\mathbb{C})$.

Now for $\sigma = (a_{ij}) \in U(n)$ if and only if

$${}^t\overline{\sigma}\sigma = \epsilon \iff \sum_k a_{ki}\overline{a_{kj}} = \delta_{ij}$$

implies $\sum_{k=1}^n |a_{ki}|^2 = 1$ which means for $i^{th}$ column vector, the absolute sum is 1.

This implies for any $1 \leq i, j \leq n$ $|a_{ij}| \leq 1$.

Now in $(C^n, ||\cdot||)$, $v = (v_k)_{k=1}^{n^2}$, $|v_k| \leq 1$ for $1 \leq k \leq n^2$ we define $||v|| = \max_{1 \leq k \leq n^2} |v_k|$.

Observe $f : M_n(C) \to C^{n^2}$ as homeomorphism $f(UC_n)$ is closed and bounded, and compact. This implies $UC_n$ is compact.

□

Note: The choice of the norm does not matter as all norms in finite dimension vector space are equivalent (Add the proof to that in the appendix).